

A rotund, Gâteaux-smooth, nowhere-LUR renorming for spaces with complemented ℓ_1

Candidate partial-result packet for arXiv:2402.13869

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Status. Candidate substantial partial result, likely valid, pending expert review. The theorem below answers Problem 3.7 of De Bernardi–Preti–Somaglia for every separable real Banach space containing a complemented copy of ℓ_1 . It does not settle the problem for arbitrary separable Banach spaces.

1 Source problem and evidence

The source is C. A. De Bernardi, A. Preti, and J. Somaglia, *A note on smooth rotund norms which are not midpoint locally uniformly rotund*, arXiv:2402.13869 [1]. Problem 3.7 on PDF page 9 asks:

Can every infinite-dimensional separable Banach space be renormed to be rotund [and Gâteaux smooth] but with no LUR point in the unit sphere?

The original page is reproduced below. The full-width crop includes the problem label, complete statement, and immediate context.

hand, it is worth to mention that by a standard application of [7, Exercise 9.20] the norm defined in Theorem 3.3 as well as the norm defined by (1) is URED. Therefore, both norms provide an affirmative answer to [10, Problem 52.3.5] which states: *Can every infinite-dimensional separable space be renormed to be URED but not LUR?*

Remark 3.6. In [4] the first and the third named authors provide, in every infinite-dimensional separable Banach space, an example of rotund Gâteaux smooth norm which satisfies the Kadec property but it is not LUR. This construction provide a positive answer to [10, Problem 52.1.1.5]. Observe that Theorem 3.3 provides an alternative solution to the same problem. On the other hand, the norm defined in [4] does not solve [10, Problem 52.1.1.3]. Indeed, if a norm is rotund and satisfies the Kadec property then it is MLUR (see [10, Corollary 408]).

The next question have been asked to us by Tommaso Russo and to the best of our knowledge it seems to be open.

Problem 3.7. *Can every infinite-dimensional separable Banach space be renormed to be rotund [and Gâteaux smooth] but with no LUR point in the unit sphere?*

In this direction, it is worth to notice that in [19] P. Morris proved that any separable Banach space containing an isomorphic copy of c_0 admits a rotund equivalent norm such that no point of the unit sphere is WMLUR. By using a result in [12] it is possible to show that if X is a polyhedral space, then it admits a C^∞ rotund equivalent norm such that no point of the unit sphere is WMLUR (see [11, Theorem 2.1]). Finally, in [11] it is proved that every C^2 norm on a polyhedral space admits no MLUR point in the unit sphere. In particular it exists, in every separable polyhedral

The source records positive results for spaces containing c_0 and for polyhedral spaces, and proves that every separable infinite-dimensional space has a rotund Gâteaux-smooth norm which is not MLUR at at least one point [1]. The result here supplies a different broad positive class and, within that class, eliminates *all* LUR points while retaining both rotundity and Gâteaux smoothness.

2 Definitions

A norm p is *rotund* (strictly convex) if equality $p(x + y) = p(x) + p(y)$ for nonzero x, y forces x and y to be positive scalar multiples. A point $x \in S_{(X,p)}$ is *LUR* if, whenever $x_k \in S_{(X,p)}$ and $p(x + x_k) \rightarrow 2$, one has $p(x - x_k) \rightarrow 0$.

A norm p is *octahedral* if for every finite-dimensional $F \subset X$ and every $\varepsilon > 0$ there is $u \in S_{(X,p)}$ such that

$$p(y + \alpha u) \geq (1 - \varepsilon)(p(y) + |\alpha|) \quad (y \in F, \alpha \in \mathbb{R}).$$

An octahedral norm has no LUR points; a short proof is included below.

3 Proof Intuition

Cobollo and Hájek constructed a Gâteaux-smooth octahedral norm N on every space with a complemented ℓ_1 , provided the space admits some Gâteaux-smooth norm [2]. Their construction does more than assert octahedrality: relative to a decomposition $X = X_0 \oplus \ell_1$ and its canonical vectors e_n , it gives a uniform asymptotic ℓ_1 estimate for $P_{n-1}x + \alpha e_n$.

Adding an ordinary *equivalent* strictly convex norm to N would charge every normalized e_n by a fixed positive amount and could destroy that estimate. Instead, we build a continuous injective strictly convex smooth norm h which is deliberately non-equivalent and satisfies $h(e_n) = 2^{-n} \rightarrow 0$. Then

$$M = N + h$$

is still equivalent because N is, is rotund because equality for M forces equality for the strictly convex summand h , and is Gâteaux smooth because both summands are. The exact Cobollo–Hájek tail estimate shows, with no compactness guesswork, that the normalized e_n remain octahedral witnesses for M . Octahedrality then supplies, at every unit vector x , unit vectors u_k for which both $M(x + u_k)$ and $M(x - u_k)$ tend to 2, ruling out LUR at x .

The unresolved part of the source problem remains substantial: spaces not containing ℓ_1 admit no octahedral renorming at all, so a full answer must use a different global mechanism.

4 The supporting tail estimate

We use the following precise feature of the norm constructed by Cobollo and Hájek, not merely their headline theorem.

Lemma 1 (Cobollo–Hájek input). *Let X admit a Gâteaux-smooth equivalent norm and a complemented subspace isomorphic to ℓ_1 . Write, after fixing an isomorphic decomposition,*

$$X = X_0 \oplus \ell_1, \quad x = x_0 + \sum_{j \geq 1} a_j e_j, \quad P_n x = x_0 + \sum_{j=1}^n a_j e_j.$$

There is an equivalent Gâteaux-smooth norm N on X such that $N(e_n) = 1$ and, for every $\eta > 0$, all sufficiently large n satisfy

$$N(P_{n-1}x + \alpha e_n) \geq (1 - \eta)(N(P_{n-1}x) + |\alpha|) \quad (x \in X, \alpha \in \mathbb{R}). \quad (1)$$

This is Theorem 1.1 together with Proposition 2.6 and the compatible unit-ball construction in [2]. The two decisive passages are reproduced here.

space X which admits such a norm is the Hardy space $H_1(D)$, as shown (using deep results from harmonic analysis) in the monograph [6] p. 120]. This space is a separable subspace of L_1 containing a complemented copy of ℓ_1 . To find such a renorming, even in the basic case $X = \ell_1$, was open. Our main result in this note is the following.

Theorem 1.1. *Let $(X, \|\cdot\|)$ be a Banach space admitting a Gâteaux smooth (equivalent) norm and having a complemented subspace isomorphic to ℓ_1 . Then X admits a renorming $\|\|\cdot\|\|$ which is simultaneously octahedral and Gâteaux smooth.*

By [12] Lemma 28], a given element $x \in S_X$ is a very smooth point if and only if x is a point of Gâteaux smoothness in the bidual X^{**} . By [20], an octahedral Gâteaux smooth norm has no points of preserved smoothness, so our main result implies, in particular, that under the assumptions of Theorem 1.1 X has a Gâteaux smooth norm without points of preserved smoothness, i.e. no very smooth points. Notice that the notion of very

$$\|\|\cdot\|\| \geq \frac{1}{\prod_{n \in \mathbb{N}} (1 + \frac{z_n + l_n}{2})} (\|\|\cdot\|\|_0 \oplus_1 \|\cdot\|_1).$$

The proof is over. $\square$

2.1 Octahedrality of the norm

Here, we will show the octahedrality of $\|\|\cdot\|\|$. Recall that a norm $\|\cdot\|$ of a Banach space X is said to be **octahedral** if for every $\varepsilon > 0$ and every finite-dimensional subspace F of X there exists $x \in S_X$ such that

$$\|y + \alpha x\| \geq (1 - \varepsilon)(\|y\| + |\alpha|)$$

for every $y \in F$ and $\alpha \in \mathbb{R}$.

The core of the idea is that, in the set B_n , the element e_n is “close” to witnessing the octahedrality of the norm for any element in B_{n-1} , and the closeness is reduced with the increasing of the n .

Proposition 2.6. *Let $\varepsilon > 0$. Then, there exists $n_0 \in \mathbb{N}$ such that, for any $n \geq n_0$,*

$$\|P_{n-1}x + \alpha e_n\|_n \geq (1 - \varepsilon)(\|P_{n-1}x\|_n + |\alpha|).$$

In particular, the norm $\|\|\cdot\|\|$ is octahedral.

5 A small smooth strictly convex norm

Lemma 2. *In the setting of Lemma 1, there exists a continuous norm h on X such that:*

1. h is strictly convex and Gâteaux smooth;
2. $h(e_n) = 2^{-n}$ for every n ;
3. for $p \in P_{n-1}X$ and $\beta \in \mathbb{R}$,

$$h(p + \beta e_n)^2 = h(p)^2 + 4^{-n}\beta^2, \quad \text{hence } h(p + \beta e_n) \geq h(p). \quad (2)$$

The norm h need not be equivalent to the original norm.

Proof. The restriction $N_0 = N|_{X_0}$ is an equivalent Gâteaux-smooth norm on the separable space X_0 . Choose a weak-star dense sequence $(\varphi_k)_{k \geq 1}$ in $B_{X_0^*}$ and put

$$q_0(x_0) = \left(\sum_{k \geq 1} 4^{-k} |\varphi_k(x_0)|^2 \right)^{1/2}.$$

Weak-star density makes (φ_k) separating, so q_0 is a norm. It is continuous and is the pullback of the Hilbert norm under the injective map $x_0 \mapsto (2^{-k} \varphi_k(x_0))_k$. Consequently q_0 is strictly convex and Gâteaux smooth away from zero. The norm

$$r_0 = N_0 + q_0$$

is equivalent and Gâteaux smooth. It is strictly convex: equality in its triangle inequality forces equality for q_0 , hence the two images in the Hilbert space are positive collinear, and injectivity gives positive collinearity in X_0 .

For $x = x_0 + \sum_{j \geq 1} a_j e_j$, define

$$h(x) = \left(r_0(x_0)^2 + \sum_{j \geq 1} 4^{-j} |a_j|^2 \right)^{1/2}. \quad (3)$$

The series term is bounded by a constant times $\|(a_j)\|_1^2$, so h is continuous on $X_0 \oplus \ell_1$; injectivity makes it a norm. It is the ℓ_2 -sum of two strictly convex norms, hence strictly convex.

For smoothness, r_0^2 is Gâteaux differentiable everywhere (with derivative zero at the origin), and the weighted quadratic form in (3) is differentiable. Away from the origin their sum is positive, so composing with the square root proves Gâteaux smoothness of h . Finally, $h(e_n) = 2^{-n}$ and (2) follows by separating the n th coordinate in the defining sum. $\square$

6 Main partial theorem

Theorem 3. *Every separable real Banach space containing a complemented copy of ℓ_1 admits an equivalent norm which is simultaneously*

1. *rotund,*
2. *Gâteaux smooth, and*
3. *octahedral.*

In particular, the unit sphere of this norm has no LUR point.

Proof. The source paper already supplies a Gâteaux-smooth equivalent norm on every separable space [1, Theorem 3.3]; hence Lemma 1 applies. Let N be its Cobollo–Hájek norm, let h be given by Lemma 2, and set

$$M(x) = N(x) + h(x).$$

Because N is equivalent and h is continuous, M is an equivalent norm. It is Gâteaux smooth as a sum of Gâteaux-smooth norms. If equality holds in the triangle inequality for M , it must hold

separately for both summands. Equality for the strictly convex norm h forces positive collinearity. Thus M is rotund.

It remains to prove octahedrality. Fix a finite-dimensional subspace $F \subset X$ and $0 < \varepsilon < 1$. Put $\eta = \varepsilon/4$. The projections P_n converge strongly to the identity and therefore uniformly on the compact M -unit sphere of F . Taking n sufficiently large, we have simultaneously

$$M((I - P_{n-1})y) \leq \eta M(y) \quad (y \in F), \quad (4)$$

$$N(P_{n-1}x + \beta e_n) \geq (1 - \eta)(N(P_{n-1}x) + |\beta|) \quad (x \in X, \beta \in \mathbb{R}), \quad (5)$$

$$\frac{1}{1 + 2^{-n}} \geq 1 - \eta. \quad (6)$$

The middle line is (1). Since

$$M(e_n) = N(e_n) + h(e_n) = 1 + 2^{-n} =: c_n, \quad u_n := e_n/c_n \in S_{(X,M)},$$

we test octahedrality with u_n .

Take $y \in F$, $\alpha \in \mathbb{R}$, put $p = P_{n-1}y$, and let $E = M(y - p) \leq \eta M(y)$. By the reverse triangle inequality, (5), and (2),

$$\begin{aligned} M(y + \alpha u_n) &\geq M(p + \alpha u_n) - E \\ &\geq (1 - \eta) \left(N(p) + \frac{|\alpha|}{c_n} \right) + h(p) - E \\ &\geq (1 - \eta) \left(M(p) + \frac{|\alpha|}{c_n} \right) - E \\ &\geq (1 - \eta) (M(y) - E + (1 - \eta)|\alpha|) - E \\ &\geq (1 - 3\eta + \eta^2)M(y) + (1 - \eta)^2|\alpha|. \end{aligned}$$

For $\eta = \varepsilon/4$, both coefficients on the last line are at least $1 - \varepsilon$. Hence

$$M(y + \alpha u_n) \geq (1 - \varepsilon)(M(y) + |\alpha|) \quad (y \in F, \alpha \in \mathbb{R}),$$

which proves octahedrality.

Finally fix $x \in S_{(X,M)}$. Applying octahedrality to $F = \text{span}\{x\}$ with errors tending to zero gives $v_k \in S_{(X,M)}$ such that

$$M(x + v_k) \longrightarrow 2, \quad M(x - v_k) \longrightarrow 2.$$

The first convergence is the LUR test at x , while the second prevents $M(x - v_k) \rightarrow 0$. Thus x is not LUR. Since x was arbitrary, the sphere has no LUR points. $\square$

7 Upgrade attempts and remaining obstruction

The following focused strengthening attempts were checked before stopping at Theorem 3.

1. **All separable spaces via octahedrality.** This cannot work: octahedral renormability is equivalent to containing ℓ_1 , as recalled in [2]. Reflexive examples such as ℓ_2 therefore require a different mechanism.
2. **Remove complementability but retain smooth octahedrality.** Cobollo–Hájek explicitly identify removal of complementability as an open extension of their theorem [2, p. 2]. Solving that extension would enlarge the present theorem to separable spaces containing an arbitrary copy of ℓ_1 , but would still not settle all separable spaces.

3. **Perturb an arbitrary octahedral norm.** A compact injective Hilbertian norm can be made small along a selected sequence, but an arbitrary octahedral norm does not automatically provide one common witness sequence with the uniform projection estimate needed above. Moreover, smooth octahedral renormability itself is unavailable without the preceding open extension. This route was not promoted.
4. **Use an equivalent strictly convex perturbation.** Its value on the N -unit vectors e_n is bounded below, so normalization creates a fixed loss and the octahedral constant need not tend to one. The non-equivalent but continuous norm h in Lemma 2 is the repair: $N + h$ remains equivalent while $h(e_n) \rightarrow 0$.

These checks leave no credible path from the present mechanism to the exact all-separable statement. Theorem 3 is therefore promoted as a partial result rather than overstated as a full solution.

8 Verification, novelty, and limitations

Proof audit. The estimate was checked at the three interfaces where a perturbation argument can fail: (i) h is a genuine injective norm but is intentionally not equivalent; (ii) $M = N + h$ is nevertheless equivalent; and (iii) the normalization factor $c_n = 1 + 2^{-n}$ and the finite-dimensional tail error are both retained explicitly. The final coefficients are $1 - 3\eta + \eta^2$ and $(1 - \eta)^2$, so $\eta = \varepsilon/4$ is sufficient. No computational experiment is used as evidence.

Bounded novelty check. On 11 August 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2402.13869 and the phrases “rotund octahedral”, “strictly convex octahedral Gâteaux smooth”, and “complemented ℓ_1 no LUR points”. External arXiv/web searches used the same close variants and also inspected arXiv:2408.03737 and arXiv:2410.18473. No source stating the simultaneous rotund–Gâteaux-smooth–octahedral conclusion for complemented- ℓ_1 spaces was located. The decisive later input [2] states smooth octahedrality, but not rotundity or the nowhere-LUR conclusion. This is a bounded search, not a claim of exhaustive novelty; novelty confidence is moderate.

Limitations and review recommendation. The argument is written for real Banach spaces, matching the real-coordinate tail inequality used in [2]. It assumes a complemented copy of ℓ_1 and does not address separable spaces without such a copy. Expert review should focus on (a) the strict convexity and differentiability of the auxiliary norm h , (b) the passage from the cited finite-stage notation to (1) on X , and (c) the uniform finite-dimensional projection estimate in (4). If these checks pass, record this as a new substantial positive class for Problem 3.7, not as a full solution.

References

- [1] C. A. De Bernardi, A. Preti, and J. Somaglia, *A note on smooth rotund norms which are not midpoint locally uniformly rotund*, arXiv:2402.13869 (2024).
- [2] Ch. Cobollo and P. Hájek, *Octahedrality and Gâteaux smoothness*, arXiv:2408.03737 (2024); J. Math. Anal. Appl. 543 (2025).